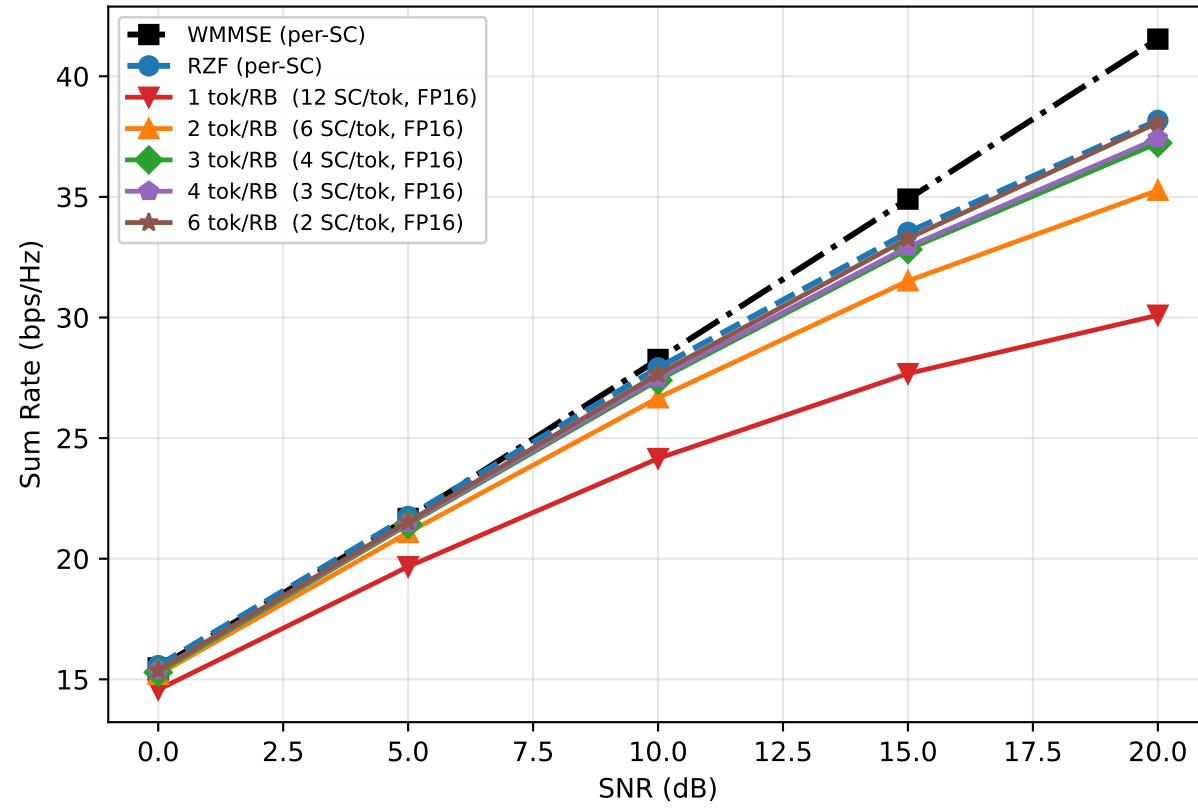


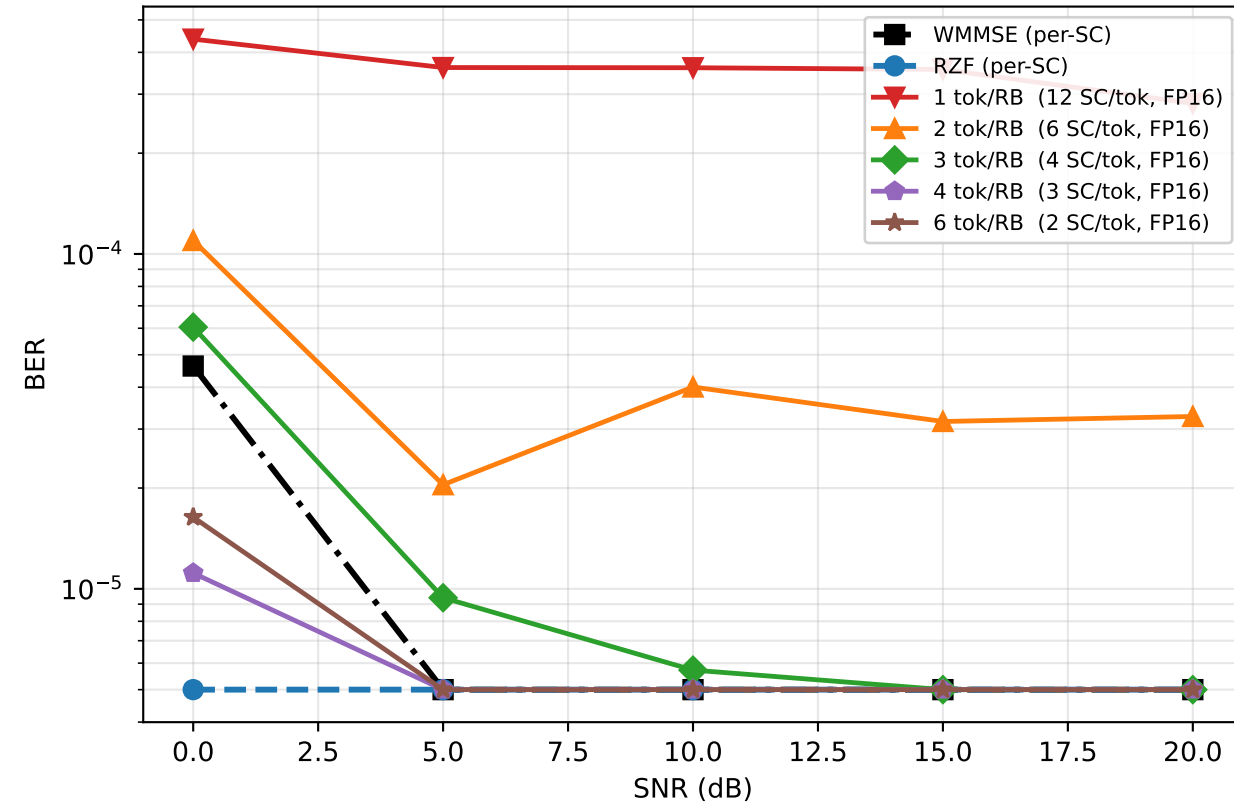
V4.2 Transformer Precoder — 8×4 MU-MIMO

8 TX antennas | 4 users | 72 subcarriers | RB = 12 SC

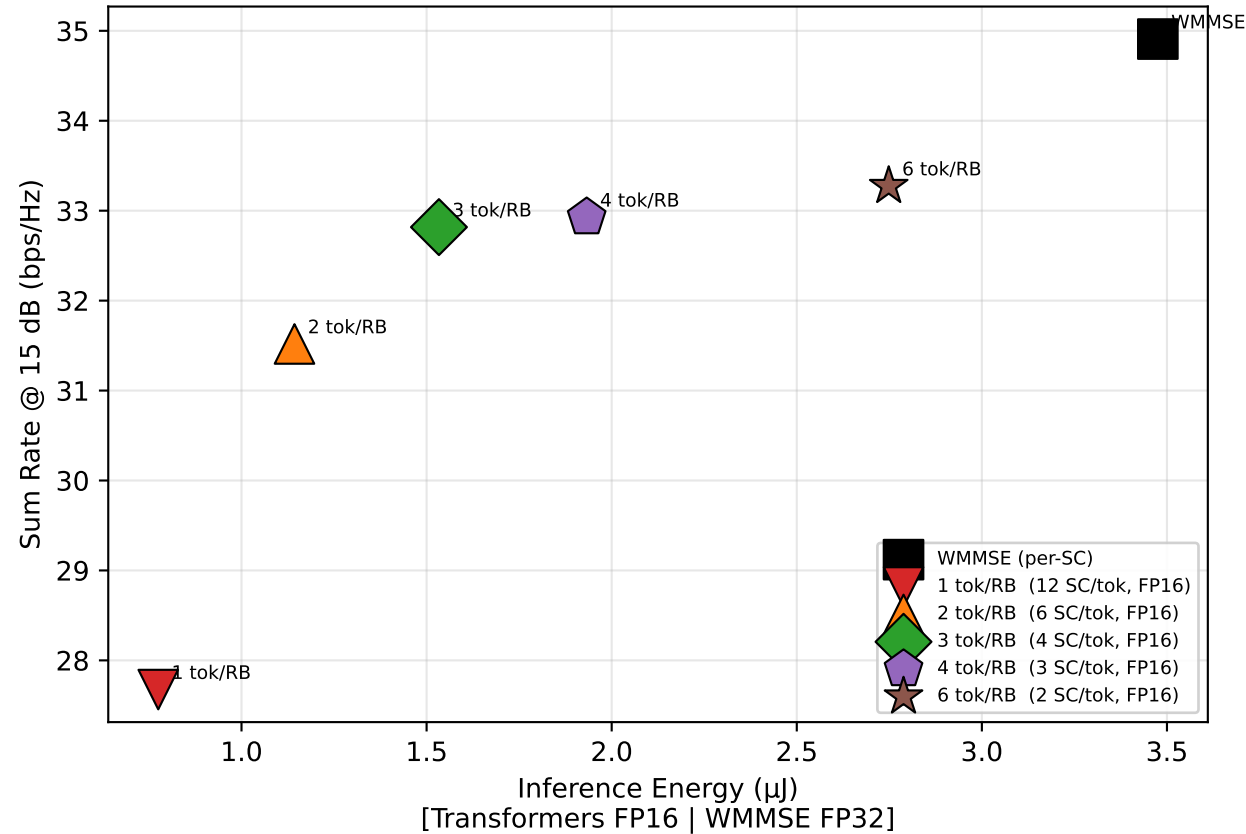
(a) Spectral Efficiency



(b) Bit Error Rate (Monte Carlo)



(c) Energy-Performance Pareto
(RZF excluded — see Fig. 2)



(d) Summary @ 15 dB
Gain = WMMSE_FP32 / model_FP16

Method	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	FP16 Energy (μ J)	Gain vs WMMSE FP32
WMMSE (per-SC)	34.91	+0.00	1842	N/A	ref
RZF (per-SC)	33.53	-1.38	2	N/A	ref
6 tok/RB (2 SC/tok, FP16)	33.27	-1.63	5080	2.7484	1.3×
4 tok/RB (3 SC/tok, FP16)	32.93	-1.98	3571	1.9325	1.8×
3 tok/RB (4 SC/tok, FP16)	32.82	-2.09	2833	1.5334	2.3×
2 tok/RB (6 SC/tok, FP16)	31.52	-3.39	2112	1.1431	3.0×
1 tok/RB (12 SC/tok, FP16)	27.68	-7.23	1432	0.7751	4.5×